

Network System Implementation for the Marconi Law Firm, LLC.

Orlando, Florida

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Table of Contents

Project Scope	2
Network Objectives/Benefits	3
IP Network Design Table	4
Network Configuration Process	5
User Logins and Passwords	7
Network Topology Diagram with Employee Names and Node Numbers	7
Capital and Operating Costs	8
Hardware and Software Specification Tables	8
Timesheets	13
Status Reports	14

Project Scope

This project is being initiated to centralize all of Marconi Law's operations into one location to enable more efficient and effective operations while simultaneously modernizing their IT infrastructure. Current communication, meetings, and day-to-day operations are being adversely affected by an outdated system and inconvenient office spaces spread across 3 separate floors. This project aims to remediate these issues by implementing a new, robust IT infrastructure on a single floor with room for scalability and growth, within five weeks.

The key objectives are to consolidate the IT infrastructure from three offices across multiple floors into a single office on one large floor while upgrading all equipment. This will enhance overall productivity and security. Objectives also include modernizing the IT infrastructure while providing ample room for scalability and growth. Specifics include:

- Providing comprehensive documentation and training for Marconi Law staff
- Designing and installing a new Local Area Network in the new office space
- Procuring and installing necessary hardware, software, printers, workstations, and networking equipment
- Ensuring data security and integrity throughout the project

Project deliverables include:

- | | | |
|----------------------|---------------------------|-----------------------|
| • 1 Server | • 1 Switch | • 10 laptops |
| • 1 Printer | • 1 Smart TV | • Windows 11 included |
| • 1 Router | • 2 WAPs | with laptops |
| • 1 Conference phone | • Windows 10 Server Setup | • 10 VoIP Phones |

This project does not include:

- | | |
|--------------------|------------|
| • Internet service | • Mice |
| • Keyboards | • Speakers |
| • Monitors | |

The project constraints include:

- **Time:** The Marconi Law firm plans to move into the new location in five weeks. This will be considered the deadline to avoid disruption to Marconi Law's operations.
- **Budget:** The project must be completed with a reasonable budget. The selected budget for this project is \$60,000.
- **Resource:** Skilled IT personnel are a necessity to ensure that the key objectives are met. The availability of skilled IT personnel and other various resources may potentially impact the project timeline.
- **Shipping:** All hardware will need to be shipped. With a deadline of five weeks, any unforeseen shipping setbacks may potentially impact the project timeline.
- **Existing Infrastructure:** Marconi Law has existing equipment. The project must integrate with some of this existing infrastructure, such as current computers and peripherals. This will ensure a smooth and successful transition.

Five key success factors that will define this project include:

- **Meeting time constraints:** The established deadline for the project is five weeks. This is when the Marconi team plans to move into the new location. This will be a strong success factor for the project
- **Staying within budget:** The established budget for the project is \$60,000. Staying within budget will help prevent extended deadlines.
- **Meeting technical specifications:** Marconi Law intends to utilize a Smart TV a conference phone, as well as a printer and various hardware and software. It is imperative that all this hardware and software meets the needs of Marconi Law. Everything needs to flow smoothly. This includes complete hardware and software compatibility.
- **Ensuring user satisfaction:** Meeting all of Marconi Law's needs while providing updated and secure IT infrastructure will ensure user satisfaction. Ongoing maintenance of the hardware and software will continue to maintain future user satisfaction.
- **Maintaining high quality:** Marconi Law is transitioning to a completely updated IT infrastructure designed specifically to meet their current needs as well as allowing room for scalability and growth. A key ingredient in maintaining user satisfaction includes maintaining the quality of their new IT infrastructure.

This project will take approximately five weeks, and cost approximately \$50,756.72.

Network Objectives/Benefits

- **Improved day-to-day operations and communication:** Marconi Law currently has several offices on multiple floors, resulting in reduced day-to-day operations and communication. This network, designed to combine all offices into one single floor, will result in an overall improvement in both aspects.
- **A modern and robust IT infrastructure:** Current equipment is outdated. This project will provide all new laptops, phones, networking equipment, as well as various other devices. The resulting benefit will be a completely overhauled and modernized IT infrastructure.
- **A secure network:** With a secure server, router, switch, WAP's, and proper operating procedures, Marconi Law will benefit greatly from this project. All devices will be secure.
- **Continued support:** The project comes with 12 months of maintenance, in which all aspects of the project will be supported and maintained. Contract renewals are welcome.

IP Network Design Table

Subnet IP Information	
Netmask	255.255.255.0
First IP	10.10.119.0
Last IP	10.10.119.255
Total Host	256
Network ID	10.10.119.0
Broadcast Address	10.10.119.255

Group IP Ranges	
Computers	10.10.119.100-10.10.119.109
VoIP Phones	10.10.119.200-10.10.119.209
Network Backbone	10.10.119.1-10.10.119.3
WAPs	10.10.119.50-10.10.119.51
Other	19.19.119.60-10.10.119.62

Node	Hardware Name	Static IP address
1	Marconi Computer	10.10.119.100
2	Jones Computer	10.10.119.101
3	Stark Computer	10.10.119.102
4	Drums Computer	10.10.119.103
5	Shoemaker Computer	10.10.119.104
6	Brooks Computer	10.10.119.105
7	Long Computer	10.10.119.106
8	Adams Computer	10.10.119.107
9	Smith Computer	10.10.119.108
10	Schultz Computer	10.10.119.109
11	Marconi VoIP Phone	10.10.119.200
12	Jones VoIP Phone	10.10.119.201
13	Stark VoIP Phone	10.10.119.202
14	Drums VoIP Phone	10.10.119.203
15	Shoemaker VoIP Phone	10.10.119.204
16	Brooks VoIP Phone	10.10.119.205
17	Long VoIP Phone	10.10.119.206
18	Adams VoIP Phone	10.10.119.207
19	Smith VoIP Phone	10.10.119.208
20	Schultz VoIP Phone	10.10.119.209
21	Router	10.10.119.1
22	Server	10.10.119.2
23	Switch	10.10.119.3
24	WAP 1	10.10.119.50
25	WAP 2	10.10.119.51
26	Printer	10.10.119.60
27	Smart TV	10.10.119.61
28	Conference Phone	10.10.119.62

Network Configuration Process

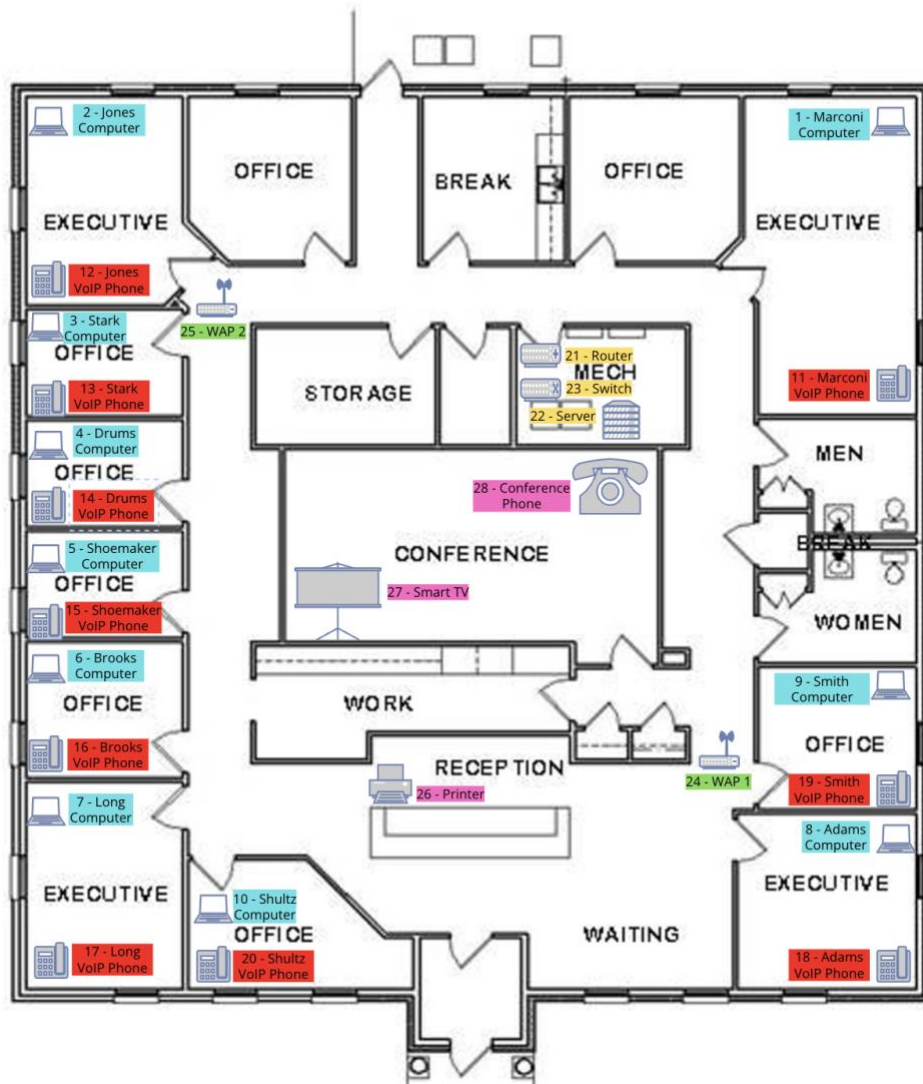
User Accounts	Setup and Configuration Process
Dan Marconi Phil Jones Karen Stark Evelyn Schultz Bill Shoemaker Mike Drums Lisa Brooks Terry Long Samuel Adams Brian Smith	Step 1: Launch Windows Server 2019 VM Step 2: Log into server using admin password Step 3: Open Server Manager by going to Windows Start Menu > Server Manager Step 4: Open Computer Management by going to Tools > Computer Management Step 5: Click Local Users and Groups in the navigation pane Step 6: Open the New User window by right-clicking Users > New User Step 7: Enter New User Username, Full Name, Password, and Confirm Password Step 8: OPTIONAL – Add user to groups for specific permissions Step 9: Uncheck User must change password at next login Step 10: Finish creating the New User by clicking Create Step 11: Test the New User Account by logging in
Groups	Setup and Configuration Process
_Attorneys _Accounting _Administrative	Step 1: Launch Windows Server 2019 VM Step 2: Log into server using admin password Step 3: Open Server Manager by going to Windows Start Menu > Server Manager Step 4: Open Computer Management by going to Tools > Computer Management Step 5: Click Local Users and Groups in the navigation pane Step 6: Open the New Group window by right-clicking Groups > New Group Step 7: Enter desired New Group name Step 8: Add desired users to New Group by clicking Add > Enter the username of the desired user > click OK Step 9: Finish creating the New Group by clicking Create

Folders and Permissions	Setup and Configuration Process
Attorneys Accounting Administrative	<u>Create Folders</u> Step 1: Launch File Explorer Step 2: Click on This PC in the navigation pane Step 3: Double-click on Local Disk (C:) Step 4: Create a New Folder by going to the menu bar, clicking Home > New Folder Step 5: Name the folder by typing “Marconi Documents” and pressing the Enter/Return key on the keyboard Step 6: Open the newly created folder by double-clicking on Marconi Documents Step 7: Add a sub-folder in Marconi Documents by going to the menu bar, clicking Home > New Folder Step 8: Name the desired sub-folder by typing the name and pressing the Enter/Return key on the keyboard <u>Set Sharing Permissions</u> Step 1: Launch Windows Server 2019 VM Step 2: Log into server using admin password Step 3: Open Server Manager by going to Windows Start Menu > Server Manager Step 4: Open Computer Management by going to Tools > Computer Management Step 5: Expand Shared Folders by double-clicking on Shared Folders in the navigation pane Step 6: Launch the New Share Wizard by right-clicking Shares > New Share Step 7: Click Next to go to the next step in the New Share Wizard Step 8: Navigate to the desired shared folder by clicking on Browse > Local Disk (C:) > Marconi Documents > select the desired folder > OK Step 9: Click Next to go to the next step in the New Share Wizard Step 10: OPTIONAL – Enter a description for the shared folder Step 11: Click Next to go to the next step in the New Share Wizard Step 12: Customize the permissions by selecting Customize Permissions > Custom Step 13: Remove universal permissions by clicking Everyone > Remove Step 14: Add Users or Groups to the custom permissions by clicking Add > Enter the Group Name or Username of the desired user > Click OK Step 15: Finalize the New Share by clicking Finish

User Logins and Passwords

Employee Name	Employee Title	Login ID	Password
Dan Marconi	President/Attorney	dmarconi	Fullsail1!
Phil Jones	Attorney	pjones	Fullsail1!
Karen Stark	Attorney	kstark	Fullsail1!
Evelyn Schultz	Administrative Assistant	eschultz	Fullsail1!
Bill Shoemaker	Attorney	bshoemaker	Fullsail1!
Mike Drums	Attorney	mdrums	Fullsail1!
Lisa Brooks	Attorney	lbrooks	Fullsail1!
Terry Long	Attorney	tlong	Fullsail1!
Samuel Adams	Attorney	sadams	Fullsail1!
Brian Smith	Accountant	bsmith	Fullsail1!

Network Topology Diagram with Employee Names and Node Numbers



Capital and Operating Costs

Hardware and Software	Per Unit Cost	Quantity	Total Cost
Laptops	\$1,239.99	10	\$12,399.90
Router	\$2,019.99	1	\$2,019.99
Server	\$4,649.00	1	\$4,649.00
Switch	\$1,748.99	1	\$1,748.99
Server Hard Drive	\$391.99	1	\$391.99
VoIP Phones	\$311.99	10	\$3,119.90
Conference Phone	\$749.00	1	\$749.00
WAPs	\$303.99	2	\$607.98
Printer	\$1,399.99	1	\$1,399.99
Smart TV	\$2,699.99	1	\$2,699.99
Windows Server 2022	\$969.99	1	\$969.99
Windows 11 Pro	N/A	N/A	N/A
Labor	\$125	40 hours	\$5,000
Annual ongoing network maintenance and support	\$15,000	1 year	\$15,000
GRAND TOTAL	-	-	\$50,756.72

Hardware and Software Specification Tables

Laptops	
Brand/Manufacturer	Lenovo
Product Line	ThinkPad T14 Gen 5
Model	21MC
Processor/Clock Speed	AMD Ryzen 7 Pro 8840U, 3.3 GHz
Memory/Speed	32 GB DDR5 SDRAM, 5600 MHz
Hard Drive/Capacity	512 GB NVMe M.2 2280 SSD
Embedded Security	TCG Opal Encryption 2
Display/Size/Resolution	14" IPS Touchscreen 1920 x 1200
Networking	Wi-Fi 6E, Gigabit Ethernet, Bluetooth 5.3
Operating System	Windows 11 Pro
Cache	16 MB L3 cache
Page URL	https://www.cdw.com/product/lenovo-thinkpad-t14-gen-5-ai-ready-14-amd-ryzen-7-pro-8840u-32-g/7978368?pfm=srh

Server	
Brand/Manufacturer	Hewlett-Packard
Product Line	HPE ProLiant
Model	DL380 Gen 11 Smart Choice
Hard Drive/Capacity	SSD 2 x 480 GB
Processor	Intel Xeon Silver 4509Y
Memory/Speed	64 GB DDR5 SDRAM, 5600 MHz
Storage Controller	HPE MR408i-o Gen11 SPDM
Monitor	None
Ethernet	4 x Gigabit Ethernet
Operating System	No OS
Networking	Broadcom BCM5719
Expansion Bays	1 (total) / 1 (free) x external Universal Media Bay, 8 (total) / 8 (free) x hot-swap 2.5" SFF
Expansion Slots	1 (total) / 1 (free) x PCIe 5.0 x16, 2 (total) / 2 (free) x FlexibleLOM
Page URL	https://www.cdw.com/product/hpe-proliant-dl380-gen11-network-choice-rack-mountable-xeon-silver-4509y/7929913?pfm=srh

Server Hard Drive	
Brand/Manufacturer	Hewlett-Packard
Product Line	HPE
Storage Type	SSD
Hard Drive Type	Hot-swap hard drive
Form Factor	2.5"
Interface	Serial ATA-600
Capacity	480 GB
Data Transfer Rate	600 MBps
Page URL	https://www.cdw.com/product/hpe-ssd-480-gb-sata-6gb-s/7768437?pfm=srh

Router	
Brand/Manufacturer	Ubiquiti
Product Line	Ubiquiti EdgeRouter Infinity
Model	ER-8-XG
RAM	16 GB DDR4 SDRAM
Flash Memory	12 GB
WAN Ports Qty	(1) GbE RJ45, or (1) 10G SFP+
Integrated Switch	None
Data Link Protocol	Up to 10 GbE
Network/Transfer Protocol	IPv4, IPv6
Page URL	https://www.cdw.com/product/ubiquiti-edgerouter-infinity-er-8-xg-router-rack-mountable/4766982?pfm=srh

Switch	
Brand/Manufacturer	Ubiquiti
Product Line	Ubiquiti UniFi Switch
Model	USW-Enterprise-48-PoE
RAM	4 GB DDR4
Flash Memory	16 GB eMMC
Performance	130.95 Mpps forwarding rate, 720W total
Capacity	176 Gbps
Ports Qty*	48-port 2.5 GbE PoE+
Page URL	https://www.cdw.com/product/ubiquiti-unifi-switch-usw-enterprise-48-poe-switch-48-ports-managed/6781041?pfm=srh

Printer	
Brand/Manufacturer	Brother International
Product Line	Brother
Model	MFC-L9670CDN
Duty Capacity	150,000 pages
Printing Output Type	Color laser
Processor	800 MHz
RAM	2 GB
Networking	Gigabit LAN, USB 2.0, USB host, Wi-Fi(n)
Page URL	https://www.cdw.com/product/brother-mfc-l9670cdn-multifunction-printer-color/7293457?pfm=srh

Wireless Access Points (WAPs)	
Brand/Manufacturer	Ubiquiti
Product Line	Ubiquiti UniFi
Model	U6
Antenna	2x2 MU-MIMO, 4x4 MU-MIMO
Capacity	10.2 GBps
Networking	2.4 GHz, 5 GHz, 6 GHz
Interface	1 x 2.5 GBase-T (PoE+) - RJ-45
Power	PoE+
Page URL	https://www.cdw.com/product/ubiquiti-unifi6-enterprise-wi-fi-6e-access-point/7193804?pfm=srh

Smart TV	
Brand/Manufacturer	Samsung
Model	QN65S90DAF
Series	S90D Series
Video Interface	HDMI
HDMI Ports Qty	4
Diagonal Size	64.5"
Resolution	3840 x 2160
Display Format	4K UHD (2160p)
Viewing Angle	Ultra viewing angle
Digital Audio Format	Dolby Atmos, Dolby Audio
USB Ports Qty	2
Wi-Fi Protocol	802.11ac
Page URL	https://www.cdw.com/product/samsung-qn65s90daf-s90d-series-65-class-64.5-viewable-oled-tv-4k/7849337?pfm=srh

Conference Phone	
Brand/Manufacturer	Grandstream
Product Line	Grandstream
Model	GAC2570
Type	LCD display conference VoIP phone
Call Services	Call Hold, Call park/pickup, Call Waiting
Speakerphone	Yes
Features	Multiple VoIP protocol support, Acoustic echo cancellation (AEC), Adaptive jitter buffer (AJB), Comfort noise generation (CNG), HD audio, Packet loss concealment (PLC), Voice activity detection (VAD)
Phone connections	1 x Ethernet 10Base-T/100Base-TX/1000Base-T
Page URL	https://www.cdw.com/product/grandstream-gac2570-conference-voip-phone-with-bluetooth-interface-12/8033336?pfm=srh

Server Software	
Brand/Manufacturer	Microsoft
Product Line	Microsoft Windows
Version	Server 2022 Standard
Licensing	License - 16 cores
Page URL	https://www.cdw.com/product/microsoft-windows-server-2022-standard-license-16-cores/6729573?pfm=srh

Client Software	
Brand/Manufacturer	License included with laptop
Product Line	N/A
Version	N/A
Licensing	N/A
Page URL	N/A

VoIP Phones	
Brand/Manufacturer	Cisco
Product Line	Cisco IP Phone
Model	7861
Type	LCD display VoIP phone
Conference Call Capability	Yes, with multiple conferencing
Call Services	Call Forwarding, Call Hold, Call Transfer, Call Waiting, Caller ID, Message Waiting Capability, Voicemail
Speakerphone	Yes
Additional Functions	Call timer, 16 lines
Features	Integrated Ethernet switch, Multiple VoIP protocol support
VoIP Protocols	SIP, SRTP
IP Address Assignment	DHCP, Static
Network Protocols	Cisco Discovery Protocol (CDP), SRTP
PoE Support	None
Voice Features	None
Page URL	https://www.cdw.com/product/cisco-ip-phone-7861-voip-phone/3179198?pfm=srh

Timesheets

Milestone	Activity Description	Start Time	End Time	Total Activity Time	Activity Date
Milestone 1	Review the case study	1:00 p	2:00 p	1 hour	1/7/2025
Milestone 1	Edit the network diagram	12:45 p	2:45 p	2 hrs	1/8/2025
Milestone 1	Select the hardware/software	2:00 p	7:00 p	5 hrs	1/8/2025
Milestone 2	Rename the Virtual Machines	7:00 a	9:00 a	2 hrs	1/14/2025
Milestone 2	Work on the project scope	1:00 p	6:00 p	5 hrs	1/15/2025
Milestone 2	Update the Network Diagram	2:00 p	5:00 p	3 hrs	1/17/2025
Milestone 3	Create the user accounts	1:00 p	4:00 p	3 hrs	1/21/2025
Milestone 3	Create groups for users	11:00 a	2:00 p	3 hrs	1/22/2025
Milestone 3	Set permissions for groups	6:00 a	9:00 a	3 hrs	1/23/2025
Milestone 4	Finalize the project report	2:00 p	6:00 p	4 hrs	1/27/2025
Milestone 4	Finalize LAN for Marconi Law	1:00 p	5:00 p	4 hrs	1/28/2025
Milestone 4	Create the pitch presentation	12:00 p	5:00 p	5 hrs	1/30/2025
TOTAL HOURS				40 hrs	

Status Reports

Accomplished this week

Milestone 1: Successfully added all hardware to the network diagram
Milestone 1: Selected all hardware while accommodating current and future needs
Milestone 1: Finalized the costs of all equipment with a grand total within budget
Milestone 2: Successfully labeled and finalized the network diagram
Milestone 2: Worked on the project scope
Milestone 2: Established static IP addresses for the proposed LAN
Milestone 3: Created user accounts for all employees
Milestone 3: Created groups to simplify management of user accounts
Milestone 3: Established appropriate security permissions for groups of users
Milestone 4: Finalized and fully formatted the project report
Milestone 4: Finalized the fully functional LAN for Marconi Law
Milestone 4: Created a pitch presentation for Marconi Law

Planned for next week

Coming up for milestone 2: Compose the project scope
Coming up for milestone 2: Establish static IP addresses for the proposed LAN
Coming up for milestone 2: Finalize the network topology diagram
Coming up for milestone 3: Create user accounts for all employees
Coming up for milestone 3: Create groups to simplify management of user accounts
Coming up for milestone 3: Establish appropriate security permissions for groups of users
Coming up for milestone 4: Finalize the fully functional LAN for Marconi Law
Coming up for milestone 4: Finalize and fully format the project report
Coming up for milestone 4: Create a pitch presentation for Marconi Law

Comments for each milestone

Milestone 1:

What problems did you run into? **I did not run into any issues**

How did you fix them?

Is there anything that you did not complete that you now have to push off into the next milestone?

Any other comments?

Milestone 2:

What problems did you run into? **I did not run into any issues**

How did you fix them?

Is there anything that you did not complete that you now have to push off into the next milestone?

Any other comments?

Milestone 3:

What problems did you run into? **I did not run into any issues**

How did you fix them?

Is there anything that you did not complete that you now have to push off into the next milestone?

Any other comments?

Milestone 4:

What problems did you run into? **I did not run into any issues**

How did you fix them?

Is there anything that you did not complete that you now have to push off into the next milestone?

Any other comments?